

Course	School/University	Year	CGPA/%
B.A. (Honors) Economics	St. Stephen's College, University of Delhi	2027	9.39/10
Grade 12 (Commerce with Applied Math)	The Indian High School, Dubai	2024	94.75% (BFS)
Grade 10	The Indian High School, Dubai	2022	94.5% (BFS)

Core: Statistics · Mathematical Methods · Microeconomics · Macroeconomics · Game Theory · Growth · Development

Electives: Advanced Econometrics · Calculus · Linear Algebra · Graph Theory · Real Analysis

PROJECTS

- **Priced In, Bit by Bit: Measuring Information Aggregation in Prediction Markets** [Paper] [Jun–Sep 2026]
 - Built a 43 million row hourly panel from Polymarket’s API (50,000 markets); cut memory usage **86.9%** (5.35GB → 699MB).
 - Applied **Shannon’s information theory** to quantify how much outcome uncertainty prices resolve; 53.65% on average, rising from 36.36% at 4+ months to over 90% one day from resolution, robust across market structures (Negative Risk vs. standard).
 - Tested predictability of price changes via **Samuelson’s martingale property**; found statistically significant but economically negligible deviations from a fair game ($R^2 = 0.025$), consistent with efficient information incorporation.
- **Income Segregation in an Agent-Based Housing Market** [GitHub] [Aug 2025–]
 - Extended **Schelling’s segregation model** with a budget-constrained rental market, with agents having **Cobb–Douglas preferences** over neighborhood income composition and residual consumption, calibrated to Delhi income microdata (CMIE).
 - Ran sealed-bid auctions across 100 neighborhoods per round with endogenous rents and displacement of priced-out tenants; tracked sorting via **Theil decomposition** into within- and between-neighborhood inequality over a **1M-agent** population.
 - Simulated a **below-market-rate unit set-aside** against a Monte Carlo baseline to measure targeting by income bracket.
- **Strategic Deception under Non-Identifiable Demand** [Paper] (Nagpaul Advisor: Prof. Anant Singh) [Nov 2025–May 2026]
 - Developed a two-period Bayesian duopoly where firms choose deception under unknown consumer tolerance, and cannot infer true market demand from observed sales due to overlapping demand mappings across regimes.
 - Showed belief heterogeneity generates a unique asymmetric BNE with strategic complementarity in deceptive marketing.
 - Proved that competition amplifies belief errors, creating a strictly positive probability of complete market collapse.
- **State Data Center Policy Competitiveness Index** (Co-author: Nathanya Eby Thomas) [Aug 2026]
 - Contributed to the empirical analysis of a 17-state policy index constructed from **state government policy documents**; conducted Monte Carlo sensitivity analysis using 20,000 **Dirichlet-distributed** dimension-weight vectors.
 - Quantified rank stability and identified **policy dimensions driving rank changes** using state-level rank distributions and dimension-sensitivity diagnostics.
- **Profits, Prices, and the Power of Information** (Co-authors: Madhura Mukund and Tanishi Singhal) [Article] [Aug 2025]
 - Published in SRCC’s *Vitta*, Annual Research Journal of the Finance and Investment Cell; **selected among 10/1,700 articles** from 5 countries for publication in **India’s oldest undergraduate finance journal**.
 - Used price theory and the **Grossman–Stiglitz paradox** to analyze the competitive race for information in algorithmic markets, examining how profit incentives drive information acquisition and shape price discovery and market efficiency.

ACHIEVEMENTS

- **Professor Nagpaul Fellow (2025): Only second-year Economics student selected** for a research fellowship by the Department of Mathematics, St. Stephen’s College; conducted research in **Game Theory with Incomplete Information**.
- **Project Euler:** Top 5.6% global rank among 925K users; solved 58 mathematical coding problems (number theory, algorithms).
- **National 2nd Runner Up – Lyons Range 15.0 (2026):** Mock stock competition held by Kirori Mal College, University of Delhi.
- **Runner Up: Interdepartmental Football, St. Stephen’s College (2026):** 2nd out of 10 teams, starting left back.
- **National Award Winner (UAE) – F1 in Schools (2023):** Won best portfolio design; managed an INR 3.7L (\$3,860) team budget.
- **Toastmasters (2024):** Ranked 1st in Table Topics (extempore) and 3rd in evaluation at the Annual Club Contest.

TECHNICAL SKILLS

- **Programming:** Python (NumPy, Pandas, Matplotlib, SciPy, SymPy, Numba), R (tidyverse, ggplot2)
- **Computational Economics:** Dynamic Programming, Markov Chains, Monte Carlo Methods; QuantEcon
- **Tools:** Git, L^AT_EX, Excel

POSITIONS OF RESPONSIBILITY

- **Junior Strategist | Indian Game Theory Society, St. Stephen’s Chapter (2025-26):** Designed the 2025–26 event roadmap; organized lectures on strategic behavior for 40+ students; designed live games and wrote articles to teach game-theoretic concepts.
- **Volunteer | Economics Society (2024-25):** Served on the editorial team for the annual journal *Towards Equilibrium*.
- **Team Leader | Weathering Heights (2023):** Led a group of 9-12th graders in building a weather station using NOAA satellites at school in collaboration with Dubai Astronomy Group, featured in leading national papers Khaleej Times and Gulf News.